

H524 Introduction to Biostatistics

Week 2 Lab: R Basics and Introduction to Probability

Fall 2025

Format: Online

Tools Required: R, RStudio

Estimated Time: 90 minutes

1 Learning Objectives

By the end of this lab, you will be able to:

1. Use basic R arithmetic and mathematical operations
2. Work with vectors and perform calculations on health data
3. Create and manipulate data frames for medical studies
4. Calculate simple probabilities using R
5. Build 2x2 tables for diagnostic testing
6. Use R functions for binomial and normal distributions

2 Part 1: R as a Calculator

Let's start with the absolute basics. R can be used as a simple calculator.

2.1 Exercise 1.1: Basic Arithmetic

Type these commands into R and observe the results:

```
# Addition
5 + 3

# Subtraction
10 - 4

# Multiplication
6 * 7

# Division
20 / 4

# Exponentiation (power)
2^3

# Order of operations (PEMDAS applies!)
(5 + 3) * 2
```

```
# Square root
sqrt(16)

# Absolute value
abs(-10)
```

Try it yourself: Calculate $\frac{120+140+135}{3}$ (the average of three blood pressure readings) using R.

2.2 Exercise 1.2: Storing Values in Variables

Instead of repeating numbers, we can store them in variables:

```
# Store a single number
age <- 45
age

# Store a calculation result
bmi_numerator <- 75 # weight in kg
bmi_denominator <- 1.7^2 # height in meters, squared
bmi <- bmi_numerator / bmi_denominator
bmi

# Round to 1 decimal place
round(bmi, 1)
```

Practice: Create variables for systolic blood pressure (120) and diastolic blood pressure (80), then calculate their average (mean arterial pressure approximation).

3 Part 2: Working with Vectors (Lists of Numbers)

In health research, we rarely work with single numbers. We usually have data from multiple patients.

3.1 Exercise 2.1: Creating Vectors

A vector is a list of values. We create vectors using the `c()` function (combine).

```
# Ages of 5 patients
ages <- c(25, 34, 45, 52, 38)
```

```
ages

# How many patients?
length(ages)

# What's the average age?
mean(ages)

# What's the oldest patient's age?
max(ages)

# What's the youngest?
min(ages)
```

3.2 Exercise 2.2: More Vector Operations

```
# Systolic blood pressure readings from 7 patients
systolic_bp <- c(120, 135, 142, 118, 160, 125, 138)

# Basic statistics
mean(systolic_bp)
median(systolic_bp)
sd(systolic_bp) # standard deviation
min(systolic_bp)
max(systolic_bp)
range(systolic_bp) # gives min and max

# Sort the values
sort(systolic_bp)

# How many readings?
length(systolic_bp)

# Sum of all readings
sum(systolic_bp)
```

Practice: Create a vector with 6 cholesterol readings: 180, 220, 195, 240, 175, 210. Find the mean, median, and standard deviation.

3.3 Exercise 2.3: Accessing Specific Values

```
# Our blood pressure data
systolic_bp <- c(120, 135, 142, 118, 160, 125, 138)
```

```
# Get the first value
systolic_bp[1]

# Get the third value
systolic_bp[3]

# Get the last value
systolic_bp[length(systolic_bp)]

# Get the first three values
systolic_bp[1:3]

# Get values in positions 2, 4, and 6
systolic_bp[c(2, 4, 6)]
```

4 Part 3: Creating Data Frames

A data frame is like a spreadsheet - it has rows and columns.

4.1 Exercise 3.1: Building a Simple Data Frame

```
# Create a small clinical trial dataset
patient_data <- data.frame(
  patient_id = c(1, 2, 3, 4, 5),
  age = c(45, 52, 38, 61, 47),
  gender = c("M", "F", "F", "M", "M"),
  systolic_bp = c(120, 135, 142, 118, 160),
  has_diabetes = c(FALSE, TRUE, FALSE, TRUE, FALSE)
)

# View the data
patient_data

# Look at the structure
str(patient_data)

# Get summary statistics
summary(patient_data)
```

4.2 Exercise 3.2: Accessing Data Frame Columns

```
# Access a column using $
patient_data$age
patient_data$systolic_bp

# Calculate mean age
mean(patient_data$age)
```

```
# Calculate mean blood pressure
mean(patient_data$systolic_bp)

# Count how many have diabetes
sum(patient_data$has_diabetes)

# What proportion have diabetes?
mean(patient_data$has_diabetes)
```

5 Part 4: Introduction to Probability in R

Now let's use R to calculate some simple probabilities!

5.1 Exercise 4.1: Simple Probability Calculations

Scenario: In a group of 100 patients, 15 have diabetes.

```
# Total patients
total_patients <- 100

# Patients with diabetes
diabetes_patients <- 15

# Probability of diabetes
prob_diabetes <- diabetes_patients / total_patients
prob_diabetes

# Convert to percentage
prob_diabetes * 100

# Probability of NOT having diabetes
prob_no_diabetes <- 1 - prob_diabetes
prob_no_diabetes
```

5.2 Exercise 4.2: Creating a 2x2 Table

Let's analyze a diagnostic test for a disease.

```
# Create a 2x2 table (matrix in R)
# Rows: Disease status (Yes/No)
# Columns: Test result (Positive/Negative)

test_results <- matrix(
  c(85, 5,      # Disease Yes: 85 test+, 5 test-
    10, 900), # Disease No: 10 test+, 900 test-
  nrow = 2,
  byrow = TRUE,
  dimnames = list(
    Disease = c("Yes", "No"),
```

```

    Test = c("Positive", "Negative")
  )
)

# View the table
test_results

# Add row and column totals
addmargins(test_results)

```

5.3 Exercise 4.3: Calculating Sensitivity and Specificity

```

# From our table above:
# True Positives (TP): disease Yes, test Positive
TP <- test_results[1, 1]

# False Negatives (FN): disease Yes, test Negative
FN <- test_results[1, 2]

# False Positives (FP): disease No, test Positive
FP <- test_results[2, 1]

# True Negatives (TN): disease No, test Negative
TN <- test_results[2, 2]

# Sensitivity: Pr(Test+ | Disease+)
sensitivity <- TP / (TP + FN)
cat("Sensitivity:", round(sensitivity * 100, 1), "%\n")

# Specificity: Pr(Test- | Disease-)
specificity <- TN / (TN + FP)
cat("Specificity:", round(specificity * 100, 1), "%\n")

# Positive Predictive Value: Pr(Disease+ | Test+)
PPV <- TP / (TP + FP)
cat("PPV:", round(PPV * 100, 1), "%\n")

# Negative Predictive Value: Pr(Disease- | Test-)
NPV <- TN / (TN + FN)
cat("NPV:", round(NPV * 100, 1), "%\n")

```

6 Part 5: Probability Distributions in R

R has built-in functions for working with probability distributions.

6.1 Exercise 5.1: Binomial Distribution

Scenario: A treatment has a 70% success rate. We treat 10 patients.

```

# Probability of exactly 7 successes
dbinom(7, size = 10, prob = 0.7)

# Probability of 7 OR MORE successes
# Method 1: Add them up
dbinom(7, 10, 0.7) + dbinom(8, 10, 0.7) +
  dbinom(9, 10, 0.7) + dbinom(10, 10, 0.7)

# Method 2: Use pbinom (much easier!)
pbinom(6, size = 10, prob = 0.7, lower.tail = FALSE)

# Expected number of successes
expected <- 10 * 0.7
expected

# Visualize the distribution
x <- 0:10 # Possible number of successes
probs <- dbinom(x, size = 10, prob = 0.7)

barplot(probs,
        names.arg = x,
        xlab = "Number of Successes",
        ylab = "Probability",
        main = "Binomial Distribution (n=10, p=0.7)",
        col = "lightblue")

```

6.2 Exercise 5.2: Normal Distribution

Scenario: Adult cholesterol levels are normally distributed with mean 200 mg/dL and SD 40 mg/dL.

```

# What proportion have cholesterol < 240 mg/dL?
pnorm(240, mean = 200, sd = 40)

# What proportion have cholesterol > 240 mg/dL?
pnorm(240, mean = 200, sd = 40, lower.tail = FALSE)

# What proportion have cholesterol between 180 and 220?
pnorm(220, 200, 40) - pnorm(180, 200, 40)

# What cholesterol level is at the 90th percentile?
qnorm(0.90, mean = 200, sd = 40)

# Visualize the distribution
x_values <- seq(100, 300, by = 1)
y_values <- dnorm(x_values, mean = 200, sd = 40)

plot(x_values, y_values,
     type = "l",
     xlab = "Cholesterol (mg/dL)",

```

```
ylab = "Density",  
main = "Normal Distribution of Cholesterol Levels",  
col = "blue",  
lwd = 2)  
  
# Add vertical line at the mean  
abline(v = 200, col = "red", lty = 2, lwd = 2)
```

7 Practice Problems

7.1 Problem 1: Blood Pressure Analysis

Seven patients have the following systolic blood pressure readings: 115, 128, 142, 135, 120, 155, 138.

- Create a vector with these values
- Calculate the mean, median, and standard deviation
- How many patients have high blood pressure (≥ 140 mmHg)?
- What proportion have high blood pressure?

7.2 Problem 2: Diagnostic Test

A rapid flu test was given to 500 patients. The results were:

- 40 patients had the flu and tested positive
 - 5 patients had the flu and tested negative
 - 30 patients didn't have the flu but tested positive
 - 425 patients didn't have the flu and tested negative
- Create a 2x2 matrix with these results
 - Calculate the sensitivity
 - Calculate the specificity
 - Calculate the PPV (positive predictive value)

7.3 Problem 3: Treatment Success

A new medication has an 80% success rate. A clinic treats 15 patients with this medication.

- a) What is the probability that exactly 12 patients respond successfully?
- b) What is the probability that at least 12 patients respond successfully?
- c) What is the expected number of successful responses?
- d) Create a barplot showing the probability distribution

7.4 Problem 4: BMI Distribution

Adult BMI is normally distributed with mean 26.5 and standard deviation 5.

- a) What proportion of adults have BMI < 25 (normal weight)?
- b) What proportion have BMI ≥ 30 (obese)?
- c) What BMI value represents the 75th percentile?
- d) Plot the normal distribution curve

8 Saving Your Work

IMPORTANT: Save your R script file!

```
# At the top of your R script, add comments with your name and date:
# Name: Your Name
# Date: Today's Date
# Lab: Week 2 - R Basics and Probability

# Save your workspace (all objects you created)
save.image("week2_lab.RData")

# To load it later:
```

```
# load("week2_lab.RData")
```